Token-Driven Research Paradigm

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Abstract

Scientific research is undergoing a deep cognitive restructuring. Protein sequences, molecular structures, genomes, meteorological fields, and even particle physics events are being encoded as discrete token sequences; the generation, evaluation, and iterative optimization of these sequences is becoming the core operational mode of research. A new paradigm of knowledge production has emerged: the Token-Driven Research Paradigm (TDRP). TDRP employs cross-modal discrete tokens as a unified representational unit, sequence generation as the core cognitive operation, human–machine cognitive coupling as the agent structure, and produces probabilistic knowledge candidates. This paper analyzes the connotation of TDRP along four dimensions (representational unification, generative cognition, probabilistic knowledge, and coupled agency), evaluates its practical manifestations across three tiers (linguistic tokens, domain-specific tokens, and fused tokens), and argues for its historical positioning as a new stage in the evolutionary sequence of scientific paradigms.

Keywords: Token-driven research paradigm; multimodal tokens; universal discrete representation; knowledge production mode; scientific cognitive restructuring; foundation models

1 Introduction: Why Tokens

1.1 The Cross-Modal Convergence of Tokenization

Over the past five years, a profound convergence has been occurring at the foundation of scientific research. AlphaFold takes amino acid sequences as token streams to predict protein three-dimensional structures [Jumper et al., 2021]; its latest version, AlphaFold 3, further extends to structure prediction of biomolecular complexes including protein–ligand and protein–nucleic acid interactions [Abramson et al., 2024]. ESM-3 processes protein sequence, structure, and function as three types of tokens in a unified multimodal manner, generating a novel fluorescent protein equivalent to simulating 500 million years of evolution [Hayes et al., 2025]. AlphaGenome extends this logic to the genomic scale, understanding regulatory logic in long DNA sequences [Google DeepMind, 2025]. The same approach extends to molecules (SMILES encoding; Weininger 1988), genomes (k-mer tokenization; Dalla-Torre et al. 2023), microscopy images (ViT patch tokens; Dosovitskiy et al. 2021), meteorological fields (spatiotemporal patch tokens; Bi et al. 2023), and even particle physics detector events [Mikuni & Nachman, 2024].

These advances point to a single fact:

> **Tokenization has become a universal cross-modal scientific representation strategy. It transforms heterogeneous, continuous scientific signals into homogeneous, discrete symbol sequences that can be processed by unified sequence modeling frameworks.**

This fact endows "token" with theoretical weight far beyond linguistics, establishing it as the core identifier of a research paradigm.

1.2 A Generalized Redefinition of Token

The generalized definition of token is as follows:

> **Token (generalized)** is the basic representational unit obtained through discrete encoding of scientific information. Data from any scientific modality (text, images, molecules, genes, time series, spatial fields), once encoded as discrete symbols from a finite vocabulary, becomes a token sequence that can be uniformly modeled, reasoned about, and generated by sequence models.

The key feature of this definition is **modality-agnosticism**: English subwords, amino acids, SMILES characters, and visual patch codebook indices differ vastly in content but are structurally isomorphic as "discretized representational units." This isomorphism yields three direct consequences: (1) **Methodological transfer**. Methods from NLP such as pretraining–finetuning, prompt engineering, and chain-of-thought reasoning can be directly transferred to molecular generation, protein design, and other domains. (2) **Knowledge fusion**. Tokens from different modalities can be concatenated into a single model for cross-modal joint reasoning (the technical basis of GPT-4o and Gemini). (3) **Generalization of generative semantics**. "Generation" is no longer limited to text but extends to protein sequences, molecular structures, experimental protocols, and all tokenizable scientific objects.

1.3 Core Thesis

The core thesis of TDRP:

> **TDRP is a mode of knowledge production: scientific cognitive activities are reorganized around the generation, evaluation, and iteration of cross-modal token sequences. Tokens simultaneously serve as universal representational units, human–machine interaction interfaces, and knowledge expression carriers. This "trinity" role establishes TDRP as an independent knowledge production paradigm.**

2 Definition and Connotation of TDRP

2.1 Formal Definition

> **Token-Driven Research Paradigm (TDRP)** is a mode of knowledge production in which the core cognitive processes of scientific research (problem formulation, knowledge synthesis, hypothesis generation, method design, data interpretation, and result articulation) are systematically reorganized around the **generation, evaluation, iteration, and integration** of **cross-modal token sequences**. In this mode, foundation models serve as unified engines for cross-modal token sequence generation and reasoning, forming structured cognitive coupling with human researchers' domain knowledge, critical judgment, and experimental verification capabilities, jointly constituting the agent of knowledge production.

Three key choices in this definition reflect the paradigmatic nature of TDRP: replacing "natural language text" with "cross-modal token sequences" to encompass non-linguistic modalities such as molecules, proteins, and images; replacing "large language model" with "foundation model" [Bommasani et al., 2021] to encompass GPT-4, Gemini, AlphaFold, ESM, and all pretrained models; and using "restructuring" to signal the paradigmatic nature of TDRP, which describes changes in the deep cognitive structure of research.

2.2 Four Dimensions of Connotation

The connotation of TDRP unfolds along four dimensions, each revealing an essential characteristic of the paradigm.

2.2.1 Dimension 1: Representational Unification

Thesis: Tokens are the unified interface for cross-modal scientific representation.

In traditional research, scientific data of different modalities are represented and processed in fundamentally different ways: text in natural language, images in pixel matrices, molecules in graph theory or coordinates, genes in base strings, and time series in numerical vectors. This representational heterogeneity constitutes the fundamental barrier to cross-modal knowledge fusion. Methods for processing text cannot be directly applied to molecules, and methods for processing images cannot be directly applied to genomes.

Tokenization offers a radical solution: **converting all modalities into a single representational form, namely discrete symbol sequences.** This conversion is not lossless (discretization inevitably involves information compression), but it yields an enormous structural advantage: once all modalities become token sequences, unified sequence modeling methods can process them all.

The implications of this representational unification for research manifest in three ways. **Computational cross-modal analogy**: traditional cross-domain analogies rely on individual scientists' intuition, but when knowledge from different domains is represented as token sequences, models can directly "discover" cross-modal structural similarities in embedding space. **Unified reasoning over multimodal evidence**: clinical problems involve medical records, imaging, and genetic data; materials problems involve literature, spectra, and crystal structures. In TDRP, these heterogeneous data are concatenated into unified token sequences for joint model reasoning. **"Linguification" of scientific data**: tokenization endows all scientific data with discrete sequential structure similar to language, granting attention mechanisms, autoregressive generation, and in-context learning cross-modal applicability. Protein sequences acquire "grammar," molecular structures acquire "syntax," and this is no longer metaphor but technical fact.

2.2.2 Dimension 2: Generative Cognition

Thesis: The core cognitive operation in TDRP is "generation," i.e., producing new token sequences based on existing token sequences.

The core cognitive operations in traditional research are diverse: observation, classification, induction, deduction, analogy, modeling, explanation... each with its own unique cognitive structure and methodological rules. TDRP provides these diverse cognitive operations with a unified computational implementation: generation.

In TDRP, diverse cognitive operations are formally unified as token sequence generation: a **hypothesis** is the generation of testable propositions from background knowledge token sequences; an **explanation** is the generation of causal narratives from data and theory token sequences; a **design** is the generation of solution descriptions from requirement specification token sequences. These operations still differ in cognitive content, but they share a single core structural feature: **all scientific cognitive activities are formally unified as mappings from given token sequences to new token sequences, with foundation models as the universal engine implementing this mapping.**

This "generative cognition" possesses several important characteristics:

(a) Explorativeness. The generation process has inherent stochasticity (controlled by sampling temperature), allowing the same input to produce multiple different outputs. In the research context, this means models can explore multiple lines of thought for the same problem,

propose multiple explanations for the same data, and design multiple plans for the same goal. This constitutes a computational form of "divergent thinking."

(b) Compositionality. Token sequence generation proceeds token by token, with each step's choice constrained by preceding tokens. This gives the generation process compositionality: new token sequences are novel combinations of existing tokens (from patterns in training data). "Innovation" in research is also largely novel combinations of existing knowledge [Arthur, 2009], creating a structural correspondence between the compositionality of generation and that of scientific innovation.

(c) Conditionality. Generation output is highly dependent on the construction of input (prompt and context). **Researchers control the generation process not by directly specifying output content, but by carefully constructing input conditions to indirectly guide the generation direction.** This control mode fundamentally differs from "directly thinking about the problem" in traditional research. It is a "metacognitive" operation requiring researchers to think not about "what is the answer" but about "how to construct input so the model is more likely to generate valuable answers."

2.2.3 Dimension 3: Probabilistic Knowledge

Thesis: TDRP produces probabilistic knowledge candidates, not deterministic knowledge assertions.

This is the most epistemologically profound characteristic of TDRP.

In the generation process of foundation models, each token selection is sampled from a conditional probability distribution:

$$P(x_t \mid x_1, x_2, \ldots, x_{t-1}; \theta) \tag{1}$$

where θ represents model parameters encoding statistical regularities learned from training data. Any scientific statement generated by the model, be it a hypothesis, an argument, or a molecular structure, is not a logical deduction's necessary result but a probabilistic "most likely continuation."

This probabilistic nature fundamentally differs from probability in traditional science:

Probability in traditional science is epistemic: we use probability to express uncertainty because our knowledge of nature is incomplete. Bayesian posterior probability reflects "our confidence in a hypothesis given evidence." Probability is a property of knowledge about the world.

Probability in TDRP is generative: it arises from the model's generation mechanism itself. The probability of model output is not "the probability that this statement is true" but "the probability of this token sequence appearing at this position in the model's statistical worldview." There is no direct correspondence between the two: a high-probability output may be wrong (because errors in training data were learned), and a low-probability output may be correct (because correct but rare knowledge is underrepresented in training data).

This distinction has important methodological implications: **TDRP outputs should not be viewed as "knowledge" but as "knowledge candidates," that is, hypotheses awaiting verification rather than confirmed facts.**

Probability also has positive aspects. One of the core activities of scientific research is "hypothesis generation," the proposing of candidate explanations to be tested. In this regard, probability is precisely an advantage: it enables models to generate diverse candidate hypotheses rather than being locked into a single chain of reasoning by deterministic logic. The value of TDRP lies not in providing "correct answers" but in **rapidly, abundantly, and diversely generating candidate hypotheses worth testing**, which are then screened and verified by human researchers and experiments.

This probabilistic feature spans all modalities: candidate molecules in molecular generation and candidate sequences in protein design are all probabilistic knowledge candidates requiring experimental verification to confirm their validity.

2.2.4 Dimension 4: Coupled Agency

Thesis: The agent of knowledge production in TDRP is neither humans alone nor models alone, but the cognitive coupling formed between humans and models through the token interface.

In TDRP, knowledge production is attributable neither entirely to human researchers nor entirely to foundation models, but occurs in their interaction. Researchers guide the model's generation direction by constructing input token sequences (prompts, background knowledge, data); models provide candidate knowledge, expand thinking space, and suggest possibilities humans might overlook through output token sequences; researchers then iteratively optimize model output through evaluation, screening, and feedback. The final scientific knowledge is a product of this iterative process.

Tokens play the role of "interaction interface" in this coupling. Human intentions are transmitted to the model through token sequences, and the model's "knowledge" is transmitted to humans through token sequences. Tokens are the sole communication channel between two fundamentally different "cognitive systems": one based on biological neural networks with consciousness and intentionality, the other based on artificial neural networks.

This coupling structure has the following characteristics:

Asymmetry and adjustability. The coupled parties have highly complementary capabilities: humans excel at problem identification, value judgment, causal reasoning, and experimental verification, while models excel at large-scale pattern recognition, cross-domain association, and diversified exploration. TDRP's effectiveness derives precisely from this complementarity. The tightness of coupling can be adjusted according to the task: exploratory phases (literature review, hypothesis brainstorming) give the model more freedom, while precision phases (data interpretation, core argumentation) require rigorous human review.

Asymmetric responsibility. Although knowledge production is coupled, **scientific responsibility always rests with human researchers**. Models are not moral agents and cannot be held responsible for scientific errors. This asymmetry is TDRP's ethical cornerstone: it establishes the irreplaceable position of human researchers in verification and quality control while delineating the ethical baseline of model use. Humans cannot invoke "generated by the model" to evade responsibility for research quality.

Specificity of cross-modal coupling. In research involving non-linguistic modalities, such as molecular design or protein engineering, the coupling interface includes not only natural language tokens but also domain-specific tokens such as molecular tokens and protein tokens. This makes the coupling more complex: researchers need to understand the model's "behavior" in these non-linguistic token spaces, and such understanding requires deep domain knowledge. A biologist using ESM to generate candidate protein sequences needs not only prompt engineering skills but deep understanding of protein structure–function relationships; without it, they cannot effectively evaluate the biological significance of model outputs.

2.3 Unified Relationship of the Four Dimensions

The four dimensions are not independently juxtaposed but mutually supporting, jointly constituting TDRP's overall connotation.

Representational unification is the infrastructure: without cross-modal tokenization, there is no unified operational interface. **Generative cognition** is the operational logic, defining the basic form of cognitive activity in TDRP. **Probabilistic knowledge** is the output characteristic, determining the epistemological status of TDRP outputs. **Coupled agency** is

the necessary requirement: because outputs are probabilistic knowledge candidates rather than deterministic facts, human verification and judgment become indispensable.

3 Practical Manifestations of TDRP

3.1 Spectrum of Practical Forms

TDRP manifests in current research practice as a spectrum, ranging from text-level applications to cross-modal scientific discovery:

Tier 1: Linguistic token-level TDRP, encompassing research activities that primarily operate on natural language tokens.

Tier 2: Domain token-level TDRP, encompassing research activities that operate on domain-specific non-linguistic tokens (molecular tokens, protein tokens, gene tokens, etc.).

Tier 3: Fused token-level TDRP, characterized by joint operation of multiple modalities of tokens within unified frameworks.

The three tiers currently coexist and interweave.

3.2 Tier 1: Linguistic Token-Level TDRP

This is the most widespread current form of TDRP. Its core operational objects are natural language tokens, and core tools are general-purpose large language models (GPT-4, Claude, Gemini, etc.).

3.2.1 Tokenization of Literature Synthesis

Traditional literature review requires researchers to read papers one by one, extract key information, build knowledge networks mentally, and output in the form of review text. The bottleneck of this process is human reading speed and working memory capacity.

In TDRP, this process is reorganized: researchers input large volumes of literature (or their abstracts and key passages) as token sequences into models, guiding the model through carefully designed prompts to generate review-type token sequences that identify consensus and disagreement, track conceptual evolution, and discover cross-domain connections. The researcher's role transforms from "paper-by-paper reader" to "review architect and output verifier."

As context windows expand (Claude's 200K tokens can accommodate approximately 150 paper abstracts), literature synthesis tokenization is progressing from "assisted summarization" to "deep synthesis." Current models still have room for improvement in latest literature coverage and grasping subtle theoretical tensions.

3.2.2 Tokenization of Hypothesis Generation

Hypothesis generation is the most epistemologically significant application of TDRP at the linguistic token level. Traditionally, scientific hypothesis formulation has been regarded as a process requiring intuition, creativity, and deep domain understanding.

TDRP reorganizes this process: researchers input research background, existing findings, and theoretical frameworks as token sequences into models, requesting generation of candidate hypothesis sets. Models "propose" new hypotheses through probabilistic recombination of tokens based on patterns learned from vast scientific literature. Any model output is a recombination of patterns in training data, but "recombination" itself can produce novelty, just as DNA recombination can produce new genotypes. The key is that models can discover pattern combinations that human researchers might overlook due to cognitive bandwidth limitations, particularly cross-domain pattern associations.

Si et al. [2024] found in a large-scale empirical study that LLM-generated research ideas match or slightly exceed human experts in novelty but remain inferior in feasibility assessment. More groundbreaking is Google DeepMind's **AI Co-Scientist** [Gottweis et al., 2025], a multi-agent system that generates and screens scientific hypotheses through tournament-style debate mechanisms. It has achieved experimentally verified successes at Stanford University (liver fibrosis drug targets) and Imperial College London (antimicrobial resistance mechanisms). Hypothesis generation tokenization is advancing from "assisted brainstorming" toward "verifiable scientific discovery."

3.2.3 Tokenization of Writing and Expression

Tokenization of academic writing is already widespread. Large numbers of researchers use LLMs to draft paper sections, polish language, and optimize argumentation structure. Beyond efficiency gains, this also changes the **cognitive function** of writing. How the cognitive process of "thinking through writing" evolves when token generation is partially outsourced to models is a research question worth tracking.

3.3 Tier 2: Domain Token-Level TDRP

This is the tier where TDRP's conceptual power is most fully demonstrated. Here, "token-driven" transcends "text generation" to reveal its true cross-modal character.

3.3.1 Token-Driven Research in Molecular Science

Tokenization of molecules. SMILES (Simplified Molecular Input Line Entry System; Weininger 1988) encodes molecular structures as linear strings. For example, benzene is encoded as "c1ccccc1" and aspirin as "CC(=O)OC1=CC=CC=C1C(=O)O." Each character is a token. SELFIES [Krenn et al., 2020] further guarantees a bijective relationship between token sequences and valid molecular structures.

Token-driven molecular discovery. In this framework, molecular discovery is reformulated as a token generation problem: given a description of target properties (in text tokens or numerical tokens), generate a molecular token sequence with those properties. MolGPT [Bagal et al., 2022], ChemGPT, and similar models are putting this approach into practice. The latest generative AI drug discovery pipelines can reduce experimental synthesis requirements from thousands of molecules to approximately 60 out of 15 million computationally designed candidates [Merk et al., 2025], dramatically improving screening efficiency. The frontier direction is developing more "physically faithful" molecular tokenization schemes, such as 3D-aware tokenization and geometric deep learning-based tokenization.

3.3.2 Token-Driven Research in Protein Science

Tokenization of proteins. Proteins naturally possess "sequence" structure. The 20 standard amino acids constitute the protein "vocabulary," with each amino acid as a token. This natural sequentiality makes proteins the most natural non-linguistic application domain for TDRP.

Token-driven protein design. Protein language models such as ESM-2 [Lin et al., 2023], pretrained on billions of protein sequences, learn the "grammar" of proteins. The latest generation, **ESM-3** [Hayes et al., 2025], achieved a major breakthrough: processing sequence, structure, and function as three types of tokens in a unified multimodal manner, generating the novel fluorescent protein esmGFP, equivalent to simulating 500 million years of evolution. Protein tokenization itself is rapidly evolving: **GeoBPE** (geometric byte pair encoding) incorporates structural information into tokenization, **evoBPE** uses evolutionary information to guide tokenization, and **BiHiTo** (biomolecular hierarchical tokenization, 2026) establishes a multi-level

token system from atoms to protein complexes. Additionally, **"Speak to a Protein"** (2025) enables interactive multimodal protein analysis dialogue.

Protein science within TDRP clearly demonstrates how "token-driven" transcends "text processing": models generate not textual descriptions about proteins, but proteins themselves. "Generation" here is the creation of scientific objects, not narratives about them.

3.3.3 Token-Driven Research in Genomics

Tokenization of genomes. Genomic sequences consist of four bases: A, T, C, and G. k-mer tokenization constitutes the genomic tokenization scheme. DNABERT [Ji et al., 2021] was pretrained on the human genome in a BERT-like manner, and Nucleotide Transformer [Dalla-Torre et al., 2023] extended this approach to multi-species genomes.

Token-driven gene function prediction and regulatory element discovery. In these models, genomic sequences are understood as a "language," with promoters, enhancers, and splice sites as its "grammatical structures." Recent advances include **ChatNT** [Richard et al., 2025], a cross-DNA, RNA, and protein multimodal conversational agent published in *Nature Machine Intelligence*; and **AlphaGenome** [Google DeepMind, 2025], which can understand regulatory logic in long DNA sequences and predict gene expression and epigenetic features.

3.3.4 Tokenization Practices in Other Domains

Materials science. CrystaLLM and related works explore tokenization and generation of crystal structures. Recent advances include **AtomGPT** (property prediction based on atomic descriptors) and **MatterGPT** (multi-property inverse materials design, generating candidate material structures based on target property combinations).

Meteorological science. Pangu-Weather [Bi et al., 2023] pioneered tokenization of meteorological fields, surpassing traditional numerical weather prediction. **GraphCast** [Lam et al., 2023] employs a GNN architecture for 10-day global forecasting. **GenCast** [Price et al., 2024] uses diffusion models to complete 15-day ensemble forecasting in under 8 minutes. In 2025, NOAA officially deployed AI weather forecasting models in operational use, marking token-driven meteorological forecasting's transition from research to operational deployment.

Particle physics. Particle physics detector events are tokenized (particle four-momenta and types encoded as tokens) for Transformer-based event classification and anomaly detection [Mikuni & Nachman, 2024].

3.4 Tier 3: Fused Token-Level TDRP

Tier 3 represents the frontier form of TDRP: tokens from multiple modalities are jointly processed within unified model frameworks.

3.4.1 Language–Molecule Fusion

Early work (MolT5, Edwards et al. 2022; Text+Chem T5, Christofidellis et al. 2023) explored joint training of natural language tokens and molecular SMILES tokens. **BioMedGPT** [Luo, Y. et al., 2023] further fuses 2D molecular graphs, protein sequences, and natural language, enabling models to accept natural language property requirements and directly output molecular token sequences. This achieves cross-modal generation from language to molecules.

3.4.2 Language–Image–Scientific Data Fusion

Multimodal foundation models (GPT-4o, Gemini) can already process text and images simultaneously. In biomedicine, **BiomedParse** and **BiomedCLIP** (Microsoft, 2024–2025) establish

multimodal foundation models for biomedical images and text, achieving joint understanding of pathology images, radiology images, and clinical text.

3.4.3 Current Limitations and Development Directions

Fused token-level TDRP remains in an early stage. Key limitations include: uneven tokenization quality across modalities (natural language most mature, molecules and proteins next); persistent difficulty in cross-modal semantic alignment; and multi-domain expertise required for verifying multimodal outputs.

3.5 Overall Assessment

TDRP is in a stage of "uneven diffusion." Linguistic token-level TDRP has permeated broadly, domain token-level TDRP has achieved remarkable progress in several pioneer domains, and fused token-level TDRP remains in proof-of-concept stage. Practice has far outpaced conceptualization. Large numbers of researchers are engaging in token-driven research, and the TDRP conceptual framework is proposed precisely to bridge this gap.

4 Paradigm Landscape: The Evolution of Scientific Cognition from Experiment to Token

The history of scientific cognition is a history of continuously evolving representation and operational logic. The recognized sequence of scientific paradigms (experimental science as the first paradigm, theoretical science as the second, computational science as the third, and data-intensive science as the fourth, per Jim Gray) records the first four stages of this evolution. Each paradigm layers upon rather than replaces its predecessors, jointly constituting the hierarchical structure of scientific cognition. TDRP represents the new stage in this sequence.

4.1 Evolutionary Logic: Representational Substrates Give Rise to Cognitive Operations

Each paradigm transition follows the same pattern: a new **representational substrate** gives rise to new **cognitive operations**.

From the first to the second paradigm, observational records were abstracted into mathematical equations, and deductive reasoning replaced inductive enumeration. From the second to the third, analytical solutions were replaced by numerical grids, and computational simulation became a new cognitive tool. From the third to the fourth, model-driven approaches were supplemented by data-driven ones, with statistical inference extracting patterns from large-scale datasets.

The transition from the fourth paradigm to TDRP continues this pattern: **data becomes tokenized, and analysis gives way to generation.** When structured data is encoded as discrete token sequences, and when the analytical operation of "discovering patterns in data" is extended to the generative operation of "generating new knowledge based on patterns," a new cognitive layer emerges.

4.2 TDRP as an Integration Layer

TDRP integrates rather than replaces the four preceding paradigms. Computational science's differential equations become physics-constrained token frameworks: PINNs and Transformer-based PDE solvers (such as FourCastNet) embed physical laws into the sequence generation process. Data science's pattern discovery becomes the training phase of foundation models,

where models learn statistical regularities in large-scale data and then externalize these regularities as new knowledge through token generation. Experimental science provides ground truth for verifying token-generated hypotheses; ESM-3's candidate proteins still require experimental validation of function, and AI Co-Scientist's hypotheses still require wet-lab confirmation.

This integrative relationship means TDRP's effectiveness is built upon the four preceding paradigms: theoretical science provides constraints, computational science provides simulation tools, data science provides training data, and experimental science provides verification mechanisms. TDRP layers upon these foundations a new cognitive stratum with tokens as the unified interface and generation as the core operation.

4.3 TDRP's Unique Contributions

Two features absent from all four preceding paradigms confer independent paradigmatic status on TDRP.

Cross-modal representational unification. In experimental, theoretical, computational, and data science, data of different modalities (text, equations, numerical values, images, molecules, genes) have always been represented and processed heterogeneously. No prior paradigm provides a unified representational substrate for these modalities. Tokenization achieves this unification: English subwords, amino acid sequences, SMILES characters, genomic k-mers, and meteorological field patches are structurally isomorphic as discrete symbol sequences, processable by unified sequence modeling frameworks.

Generative cognition as the core operation. The four preceding paradigms are essentially analytical: observing phenomena, deriving equations, computing simulations, extracting patterns. TDRP is essentially generative. It generates new token sequences that can be new hypotheses, new molecules, new proteins, new arguments. Analytical paradigms answer "what exists in existing data"; TDRP answers "what can be generated based on existing knowledge."

4.4 Overview of Paradigm Evolution

Paradigm	Repr. Substrate	Core Operation	TDRP Inheritance
1st: Experimental	Observations	Induction	Experiments verify hypotheses
2nd: Theoretical	Equations	Deduction	Physics constraints in generation
3rd: Computational	Numerical grids	Simulation	Transformers replace solvers
4th: Data Science	Structured data	Statistical inference	Data trains foundation models
TDRP	**Cross-modal tokens**	**Generation**	**Unified repr. + generation**

Table 1: The evolutionary sequence of scientific paradigms.

5 Evolutionary Logic and Future Prospects

5.1 The Internal Logic of Evolution

TDRP's evolution is driven by an internal logic that can be summarized as the **"extension and deepening" double helix of tokenization**:

Extension: Increasingly more scientific modalities are tokenized and brought into TDRP's operational scope. From natural language → molecules → proteins → genomes → materials → meteorology → ... each new modality's tokenization expands TDRP's domain of applicability.

Deepening: Within each already-tokenized modality, the depth of token operations continuously increases, progressing from surface-level generation (e.g., text polishing) to deep cognition (e.g., hypothesis reasoning), and from single-modal processing to cross-modal fusion reasoning.

5.2 Three-Stage Evolutionary Roadmap

Based on the double helix logic, TDRP's evolution can be divided into three stages:

5.2.1 Stage 1: Text-Dominated Assisted Enhancement

Core features: TDRP primarily operates at the linguistic token level. Foundation models serve as "cognitive enhancers" embedded in research workflows, improving efficiency in text-intensive tasks such as literature synthesis, hypothesis brainstorming, and writing.

Landmark achievements (mostly realized):

- LLM-assisted literature review and paper writing have become research norms [Van Noorden & Perkel, 2023]
- ESM-3 generates novel functional proteins [Hayes et al., 2025]
- Generative AI drug discovery enters practical pipelines; AI Co-Scientist produces experimentally verified hypotheses [Gottweis et al., 2025]
- AI weather forecasting enters operational deployment (NOAA, 2025)
- Multiple surveys systematically chart LLM evolution in scientific discovery [Lu et al., 2025, Ye et al., 2025]

5.2.2 Stage 2: Cross-Modal Deep Symbiosis

Core features: TDRP extends to multiple scientific modalities with deep fusion; models upgrade from "efficiency tools" to "co-participants" in cognitive processes. Key trends include: (a) maturation of "scientific foundation models" with cross-modal scientific reasoning capabilities; (b) cross-modal hypothesis generation combining evidence from different modalities; (c) deepening human–machine "scientific dialogue" from instruction–execution to critical dialogue.

Key challenge: The imbalance between increasing model capability and lagging human verification capacity. Developing efficient verification tools, including "adversarial verification" (using models to audit models), will be the core challenge.

5.2.3 Stage 3: Autonomous Multimodal Discovery

Stage 3 is a logical extrapolation rather than a definite prediction; whether and when its realization conditions (particularly reliable model reasoning capability) can be met remains highly uncertain.

Features: Models form closed loops with automated laboratories: generating hypotheses and experimental protocols (token sequences), executing experiments automatically, tokenizing and feeding back data, then initiating a new generation cycle. Human roles shift to strategic planners and ultimate arbiters. Models directly generate scientific objects themselves (new molecules, new proteins, new materials) rather than merely descriptive text, dramatically expanding the scientific search space.

5.3 Key Conditions and Risks of Evolution

5.3.1 Key Conditions from Stage 1 to Stage 2

(1) High-quality tokenization of more scientific modalities. Effective tokenization schemes must be developed for more scientific data types. The goal is not merely discretizing data, but ensuring the tokenization process preserves scientifically important information.

(2) Technical breakthroughs in cross-modal alignment. The "semantic alignment" problem between tokens of different modalities must be solved, ensuring that concepts in language token space accurately correspond to features in molecular token space.

(3) Methodological consensus in the scientific community. Consensus must be formed on TDRP usage norms, verification standards, and transparency requirements. Without methodological consensus, TDRP outputs cannot be trusted by the scientific community.

(4) Development of verification tools. Efficient, modality-specific verification tools must be developed, including automated fact-checking, logical consistency detection, and cross-modal consistency verification.

5.3.2 Core Risks

These conditions correspond to several issues requiring sustained attention: cross-modal hallucination propagation, research direction homogenization based on shared models, imbalance between generation and verification capabilities, and long-term cultivation of independent scientific thinking abilities. These are concomitant challenges of any powerful new methodology, demanding that the scientific community maintain methodological vigilance while embracing TDRP.

5.4 A Fundamental Open Question

The ultimate trajectory of TDRP's evolution depends on a fundamental question that cannot currently be answered:

> **Can token sequence operations, no matter how complex and large-scale, exhaust the entirety of scientific cognition? Or are there scientific cognitive activities that are inherently non-tokenizable?**

If all scientific cognition can ultimately be represented as token operations, then TDRP's evolution has no principled boundary, and it will evolve from "one paradigm among many" to "the only paradigm."

But if non-tokenizable scientific cognition exists (direct intuitive experience of nature, the "feel" of experimental phenomena, the sense of scientific beauty), then TDRP will always coexist with other paradigms, and those non-tokenizable cognitive activities will demarcate TDRP's permanent boundary.

Regardless of the ultimate answer to this question, TDRP is already reshaping the core processes of scientific cognition. This fact itself establishes its paradigmatic status.

6 Conclusion

6.1 Core Contributions

The TDRP conceptual framework advances understanding on three levels. The generalization of the token concept endows "token-driven" with genuine paradigm-level theoretical weight. It is not a synonym for "using LLMs to write papers" but a conceptual identifier for "restructuring scientific cognition with cross-modal discrete representation as a unified interface." Representational unification, generative cognition, probabilistic knowledge, and coupled agency constitute TDRP's four-dimensional connotation system with internal logical relationships: representational unification provides infrastructure, generative cognition defines operational logic, probabilistic knowledge characterizes output properties, and coupled agency establishes the responsibility structure. The three-tier analytical framework (linguistic/domain/fused tokens) assesses TDRP's current progress, while the three-stage roadmap (assisted enhancement/deep symbiosis/autonomous discovery) charts evolutionary prospects and identifies key conditions and risks at each stage.

6.2 Restated Core Assertion

The core assertion, condensed:

> The Token-Driven Research Paradigm (TDRP) is an emerging mode of knowledge production. Its essence lies not in the use of any specific AI tool, but in a deeper structural transformation: when all scientific data, from paper text to molecular structures, from gene sequences to microscopy images, can be encoded as discrete token sequences, and when the generation, evaluation, and iterative optimization of these token sequences become the basic operational mode of scientific cognition, the underlying logic of scientific research is restructured. This restructuring does not replace existing paradigms but layers upon them a new cognitive stratum, one with tokens as the unified interface, generation as the core operation, and human–machine coupling as the agent form. The expansion and deepening of this cognitive layer is proceeding at a pace exceeding expectations.

6.3 To the Scientific Community

TDRP is not an option one can choose to "accept or not." It is already happening. The question is not "whether to adopt TDRP" but "how to ensure TDRP's development benefits scientific progress."

This requires the scientific community to act simultaneously on three levels:

Conceptual level: Developing clear understanding of TDRP. The TDRP conceptual framework provides a starting point for this understanding. More systematic thinking from philosophy of science, cognitive science, sociology of science, and domain disciplines is needed.

Normative level: Establishing methodological standards and ethical guidelines for TDRP. This includes reproducibility standards for token operations, minimum verification requirements, transparency disclosure norms, and clear rules for human–machine contribution attribution.

Educational level: Reforming research training systems so future researchers can both effectively engage in cognitive coupling with foundation models and maintain independent thinking and critical judgment capabilities. The balance between these two will determine whether TDRP becomes an accelerator of scientific progress or an eroder of scientific reliability.

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The writing process of this paper itself is an instance of TDRP. The significance of tokens extends far beyond language, and deeper discussions of TDRP will ultimately take forms transcending the pure text paper.

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